

LinkHealth Intake Triage

An AI triage system for inbound business enquiries

"LinkHealth" is a fictional company name used for this case study.

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- A professional-services firm receives **40–60 enquiries/week** via a web form
- Today: a junior analyst reads each submission, tags service line, estimates complexity, routes to the right lead — **~8 hours/week, error-prone**
- This system automates: **classify → score → guardrail → route**

Architecture — hub & spoke, with a hard guardrail

```
enquiry (raw_text · industry · org_size · urgency)
  → intake-triage hub
    1. classify → one of 3 service lines
    2. score complexity (4 dimensions × 0-2 → simple/moderate/complex)
    3. PHI guardrail: data_sensitivity ≥ 1 → phi_involved → requires_human_review
    4. ambiguity → needs_manual_triage (never invent a 4th category)
    5. log decision to data/triage_log.jsonl
    6. dispatch to the matching spoke
  → spoke (automation / data / deployment lead) → scoping note + questions
```

- **Two enforcement layers:** prompt instruction (SKILL.md) + deterministic backstop in code (post-write validator — the model cannot skip it)
- **The one hard stop:** any PHI-involved enquiry is queued for human review, never auto-dispatched
- **Model/API:** LLM classification via API — the same prompt assets run on Claude Code and DeepSeek Harness, so the model choice is swappable, not baked in

Test 1 · Full flow (automatic, no guardrail)

Enquiry: small dental practice, no-show revenue loss, wants reminders + prediction

Step	Result
Classification	Process & Workflow Automation
Complexity	simple (1/8 — single scheduling-app integration, admin-only data)
Guardrail	phi=false → auto-handoff permitted
Logged	data/triage_log.jsonl, validator PASS
Handoff	automation-lead produced scoping note + 3 clarifying questions

Demonstrates: the fully automatic loop, end to end.

Test 2 · Guardrail trip (PHI → human review)

Enquiry: regional hospital network, prior-authorization re-keying from EHR

Step	Result
Classification	Process & Workflow Automation
Complexity	complex (6/8 — multi-system integration, 450-bed org scale)
Guardrail	data_sensitivity=2 → phi=true → requires_human_review=true
Dispatch	BLOCKED — no spoke, queued for human compliance sign-off
Logged	validator PASS (guardrail constraint satisfied)

Demonstrates: the system's only hard stop — PHI never auto-dispatches.

Test 3 · Hard block (backstop rejects a violating write)

Attempt: append a record with `phi_involved: true` but `requires_human_review: false`

- The `tools/post-execute` backstop re-reads the file and **blocks the write**:
`[guardrail] BLOCKED: triage record ... PHI-flagged enquiries must always require human review`
- Model cannot proceed until the record is corrected; validator then returns PASS

Demonstrates: enforcement in code, not just instruction — the model cannot silently skip the guardrail.

Production thinking — what breaks first, what to monitor

Question	Answer
What breaks first?	model scoring drift on edge cases (ambiguous enquiries); log write contention at batch volume
What to monitor?	trriage_log.jsonl (append-only audit trail), guardrail block rate, per-class accuracy vs <code>expected_*</code> labels
Fallback when the model is wrong?	the deterministic backstop rejects violations; <code>needs_manual_triage</code> routes weak fits to a human; every decision is logged and replayable
Scale trigger	a real queue replaces direct dispatch once volume/multi-lead claiming demands it (PRD §7)

Extension · Deployed to a cloud VM via CI/CD

Beyond the core brief — an optional stretch showing the same capability running in production; the assignment itself is fully covered by slides 1–6.

```
git push main → GitHub Actions → build self-contained release → GCP VM  
→ releases/<sha> → current → systemd → health check  
(rollback = symlink switch)
```

- **One DSH install, two profiles:** `web` (clean dev) vs `linkhealth` (the product)
- **The profile is the deployment unit:** same files run locally and on the VM
- Private access via SSH tunnel (no public app port); synthetic data only
- Front door: brand theme + capability launcher (Triage enabled; CDI Audit ready)

Summary

- **Judgement, not boilerplate:** a deliberately small scope done deeply — classification + scoring + routing, with a guardrail designed as the core asset
- **The guardrail is the differentiator:** instruction *and* code enforcement, PHI as a hard stop, audit trail on every decision
- **Production-minded:** what breaks, what to monitor, fallback when the model errs
- **Beyond the brief:** the same capability runs as plugins on Claude Code and DeepSeek Harness, and deploys to a cloud VM through CI/CD

AI assists · a human decides · everything is auditable.